International Journal of Mental Health and Addiction
https://doi.org/10.1007/s11469-025-01502-5

ORIGINAL ARTICLE

A Cross-lagged Panel Network Analysis of the Relationship Between Emotion Regulation Difficulties and Problematic TikTok Use

Nisha Yao[1] · Jing Chen[2] · Jon D. Elhai[3,4]

Accepted: 12 May 2025
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract
The current study examined the relationships between dimensions of difficulties in emotion regulation (DER) and symptoms of problematic TikTok use (PTU) to identify key cross-construct influences and their directionality over time. Repeated measures of DER and PTU were administered to a sample of TikTok users over a 2-month period. Data from a total of 715 participants ($\text{Mean}_{\text{age}} = 27.92$, $SD = 6.01$; 65.2% females) were used in the analysis. Cross-lagged panel network analysis revealed that difficulties in impulse control and goal-directed behavior were the most influential factors exerting cross-construct effects. Bidirectional relationships, particularly between goal-directed behavior difficulties and task interference from TikTok use, were also observed. The current findings refine our understanding of how DER and PTU relate over time, which could inform the development of more precise intervention strategies to effectively reduce problematic use.

Keywords Emotion dysregulation · Problematic TikTok use · Cross-lagged panel network model · Directionality · Bridge nodes

Introduction

Watching and editing short videos have become integral to modern digital life. Among short-video platforms, TikTok (known as Douyin in China) stands out as one of the most influential, with over 1.8 billion monthly active users as of February 2023 (Aslam, 2024;

✉ Nisha Yao
yaonisha0903@outlook.com

✉ Jing Chen
chenjingps@gmail.com

Jon D. Elhai
contact@jon-elhai.com

1 School of Kinesiology and Health, Capital University of Physical Education and Sports, Beijing, China

2 School of Psychology, Shanghai University of Sport, Shanghai, China

3 Department of Psychology, University of Toledo, Toledo, OH, USA

4 Department of Neuosciences and Psychiatry, University of Toledo, Toledo, OH, USA

Published online: 04 June 2025

Montag et al., 2021). While TikTok provides entertainment and social connection, excessive use may result in adverse life consequences (Nong et al., 2023;Ye, Wu et al., 2023b; Ye, He et al., 2023a; Ye, Chen et al., 2024a; Ye, Cui et al., 2024). Problematic TikTok use (PTU), a specific form of problematic Internet use (PIU), is characterized by withdrawal-like symptoms, mood modification, preoccupation, difficulty reducing use, increased tolerance, and significant disruptions to daily life (Smith & Short, 2022). Understanding the mechanisms underlying PTU is crucial for refining theoretical models and developing effective intervention strategies.

Difficulty in emotion regulation (DER) has been conceptualized as a multidimensional construct, including the inability to understand emotions, lack of acceptance of emotions, difficulties engaging in goal-directed behavior when distressed, impulsive control difficulties, and inability to access effective emotion regulation strategies (Bjureberg et al., 2016; Gratz & Roemer, 2004). Dysfunction in one or more of these dimensions may indicate emotion dysregulation.

Theoretical models have highlighted the role of DER in the development and maintenance of PIU, providing a foundation for understanding PTU. Specifically, compensatory Internet use theory (CIUT) posits that individuals experiencing psychosocial difficulties may engage in Internet use as a means of escaping or alleviating distress (Kardefelt-Winther, 2014). Although this behavior provides temporary relief, it can lead to excessive reliance on Internet use over time, maintaining or exacerbating the underlying difficulties. In this context, individuals with DER may turn to TikTok use to manage emotional distress, yet this maladaptive coping strategy may increase vulnerability to developing PTU. The I-PACE model (Brand et al., 2016, 2019) complements this perspective by emphasizing the role of affective and cognitive responses in the development and maintenance of PIU. It suggests that DER heightens the urge to use/overuse Internet applications, such as TikTok, as a means of regulating negative mood. Importantly, this model also suggests the potential for a bidirectional effect, where prolonged reliance on Internet use for emotion regulation may impair adaptive regulation skills. This creates a maladaptive cycle in which DER and PIU reinforce one another. Together, these models provide valuable insights into mechanisms linking DER and PTU.

While extensive research has explored DER and PIU, research specifically examining DER and PTU remains scarce (Wang et al., 2023; Yang et al., 2022a, 2022b). Therefore, we first review existing findings on DER and PIU, providing a broader context for understanding how DER is linked to the problematic use of specific Internet applications like TikTok. We then examine the association between DER and PTU while addressing limitations in the existing literature.

Consistent with these theoretical perspectives (Brand et al., 2016, 2019; Kardefelt-Winther, 2014), many studies have supported the link between DER and PIU severity, with some focusing on overall DER. Cross-sectional studies have primarily examined the influence of DER on PIU and found supporting evidence (e.g., Akbari et al., 2017; Casale et al., 2016; Faghani et al., 2020; Gioia et al., 2021; Yang et al., 2022a, 2022b). Longitudinal studies further corroborate this connection. For example, in a German cultural context, one study demonstrated that baseline DER predicted PIU at follow-up in children (Paulus et al., 2021), while two others examined changes in PIU over time. One study found that higher DER was associated with increasing PIU, as compared to persistently low PIU, in children and adolescents (Schettler et al., 2024), while another reported that greater DER was linked to persistently high PIU, as compared to decreasing PIU, in adolescents (Wartberg & Lindenberg, 2020). Additionally, a cross-lagged longitudinal study accounting for parenting style showed that DER predicted later PIU severity, rather than the reverse, in a Chinese adolescent sample (Wang et al., 2023).

Other studies have examined the relationship between specific facets of DER and PIU. Cross-sectional evidence suggests that DER strategies, impulse control, goal-directed behavior, emotional clarity, and emotional acceptance are associated with PIU, while lack of emotional awareness has received less support (e.g., Amendola et al., 2019; Hew & Chew, 2025; Horwood & Anglim, 2021; Wartberg et al., 2021). However, longitudinal findings remain inconsistent. Two studies using different analytical approaches produced divergent results: Tsai et al. (2020) categorized Chinese young adults into PIU trajectory groups and found that baseline impulse control difficulties predicted increasing PIU, as compared to persistently low PIU, but not vice versa, whereas Donald et al. (2022) used a cross-lagged approach in Australian adolescents and found that PIU severity predicted later difficulties with goal-directed behavior and emotional clarity, but not the reverse. These discrepancies likely reflect differences in methodology, age groups, and cultural contexts.

Overall, most studies have examined a unidirectional pathway from DER to PIU (e.g., Gioia et al., 2021; Paulus et al., 2021; Schettler et al., 2024; Wartberg & Lindenberg, 2020), while fewer have explored the reverse pathway or considered bidirectional relationships (Donald et al., 2022; Tsai et al., 2020; Wang et al., 2023). Although findings predominantly support the influence of DER on PIU, emerging evidence suggests that PIU may also exacerbate DER. These findings align with theoretical models of PIU (Brand et al., 2016, 2019; Kardefelt-Winther, 2014), which primarily propose that DER is a risk factor for PIU, while also suggesting that PIU may perpetuate DER.

Despite these valuable insights, the existing literature is not without limitations. First, longitudinal studies examining the relationship between DER and PIU remain scarce. Of the studies reviewed, only six employed longitudinal designs (Donald et al., 2022; Paulus et al., 2021; Schettler et al., 2024; Tsai et al., 2020; Wang et al., 2023; Wartberg & Lindenberg, 2020). Although cross-sectional research consistently highlights associations between DER and PIU, longitudinal designs are essential for establishing temporal precedence and directionality, offering deeper insights into the causal pathways between these constructs. Second, the exploration of bidirectional relationships between DER and PIU has been limited. Only three of the six longitudinal studies investigated bidirectional effects, with inconsistent findings likely due to variations in analytical methods, sample characteristics, and cultural contexts (Donald et al., 2022; Tsai et al., 2020; Wang et al., 2023). Without accounting for bidirectional influences, research risks overlooking the possibility that PIU may also contribute to DER, creating a maladaptive feedback loop (Brand et al., 2016). Third, prior research has largely overlooked the multidimensional structure of both DER and PIU. Existing longitudinal studies have treated PIU as a unidimensional construct, while two studies (Donald et al., 2022; Tsai et al., 2020) explored how DER facets relate to PIU over time but did so in isolation. Donald et al. (2022) constructed separate cross-lagged models for each DER facet's relationship with PIU, whereas Tsai et al. (2020) investigated whether individual DER facets predicted changes in PIU trajectories. By analyzing DER facets separately and treating PIU as a unidimensional construct, these studies risk neglecting the interconnections among DER dimensions and the heterogeneity across PIU symptoms. Recognizing these complexities, future research should adopt more nuanced approaches to capture the interplay between DER facets and PIU symptoms, which could ultimately refine theoretical models and guide the development of targeted interventions.

The Present Study

Building on this literature, the current study aims to refine the understanding of the association between DER and PTU while addressing the above-mentioned gaps. To achieve this goal, the current study employed a cross-lagged panel network (CLPN) approach to examine how specific DER facets and PTU symptoms relate to each other over time. CLPN builds on network theory, which conceptualizes psychological constructs like DER and PIU as systems of interconnected nodes—in this case, DER facets and PTU symptoms—linked by edges that represent their relationships (Borsboom, 2017). By identifying bridge edges between DER and PTU networks, we can clarify pathways linking these constructs, while pinpointing bridge nodes allows us to determine the most central DER facets and PTU symptoms in this association. Furthermore, CLPN tracks how these connections evolve over time, enabling us to assess their directionality and clarify whether DER influences PTU, vice versa, or both.

In this study, we collected data from 715 TikTok users over a 2-month period using a repeated-measures design. First, we examined contemporaneous networks at Time 1 and Time 2 to identify the most influential bridge nodes and edges connecting DER and PTU (Epskamp et al., 2018; Jones et al., 2021). Then, applying CLPN, we analyzed how these bridge edges evolved over time and identified bridge nodes driving these temporal associations.

These analyses address two key questions: (1) What are the key bridge nodes and edges connecting DER and PTU? (2) What is the directionality of influence between them? For the first question, prior longitudinal studies in populations with similar age groups and cultural contexts suggest that impulse control difficulties may play a central role in linking DER to PIU (Tsai et al., 2020). However, existing longitudinal studies have treated PIU as a unitary construct while examining DER facets individually, limiting our ability to formulate specific hypotheses about how DER facets and PTU symptoms relate over time. Beyond identifying impulse control difficulties as a key bridge node, this study takes an exploratory approach to examining other DER facets, PTU symptoms, and their specific pathways of influence. For the second question, theoretical models and prior findings from longitudinal studies predominantly suggest that DER increases vulnerability to PIU (Paulus et al., 2021; Schettler et al., 2024; Tsai et al., 2020; Wartberg & Lindenberg, 2020), which is also evident in a study specifically examining DER and PTU (Wang et al., 2023). However, some evidence also indicates that PIU may exacerbate DER (Donald et al., 2022). Based on this, we hypothesize that the primary direction of influence flows from DER to PTU, though bidirectional effects may exist, with PTU influencing DER to a lesser extent. Addressing these questions holds both theoretical and clinical significance. The findings will help refine theoretical models of PIU and DER while informing targeted interventions and prevention strategies.

Methods

Participants

The data used in this study are from a larger dataset collected to examine processes leading to PTU. Participants were recruited through an online survey service provider Credamo

(https://www.credamo.com), which operates similarly to Amazon's Mechanical Turk. When the current study was conducted (May–August 2022), Credamo had a nationwide sample pool of around 2.8 million participants in China. According to the platform-level demographic data from May 2022, 47% of participants were female. The age distribution was as follows: 60% were 18–25 years, 29% were 26–35 years, 7% were 36–45 years, and 4% were 46 years and older. In terms of education, 54% had a college education, 40% had a high school education or below, and 6% had a master's degree or higher. Monthly income distribution showed that 31% earned 3000 Chinese Yuan (CNY) and lower, 24% earned between 3001 and 8000 CNY, 27% earned between 8001 and 15,000 CNY, and 18% earned more than 15,000 CNY. The most common occupations were students (31%); general staff working in offices (11%); professionals such as doctors, lawyers, journalists, and teachers (9%); and general workers (e.g., factory workers, manual laborers; 9%). These platform-level demographics provide useful context for understanding the broader characteristics of Credamo participants.

To identify regular TikTok users, we screened 2500 participants using three criteria: (1) TikTok had to be their primary short-form video platform, (2) they had to have a TikTok account, and (3) they had to use TikTok several times per week. A total of 1428 participants met these criteria and were invited to participate in the study. The data were collected in two waves. Participants completed an online battery of Chinese-language questionnaires assessing TikTok usage behaviors and related psychological factors. To ensure data quality, participants could only complete questionnaires once per assessment occasion. Additionally, three attention-check items (e.g., "Select the third option") were included in the questionnaires at each time point, and only participants who passed all attention checks were included in final analyses. No specific demographic variables were controlled during recruitment beyond these eligibility criteria, and participants were recruited based on platform availability.

Informed consent was obtained prior to participation, with participants receiving detailed information about the study before proceeding. Ethical approval was granted by the local ethics committee of a university in Beijing, China (2022A21). Participants received monetary compensation for their participation.

A total of 822 valid participants completed assessment at Time 1, and 715 of those participants completed it at Time 2. Only participants with complete data at both time points ($N = 715$) were included in analyses. The final sample included 466 females (65.2%), with a mean age of 27.92 years ($SD = 6.01$, range: 18–56 years). In terms of education, 610 participants (85.3%) had a college degree, 18 participants (2.5%) had a high school education or below, and 87 participants (12.2%) had a master's degree or higher. Regarding TikTok usage, on a rating scale from 1 (never) to 5 (very often), participants reported that they often or very often watched short videos ($M = 4.45$, $SD = 0.68$), while they rarely or sometimes created short videos ($M = 2.55$, $SD = 0.83$).

Measures

Difficulties in Emotion Regulation

We assessed DER using the Difficulties in Emotion Regulation Scale–16 (DERS-16; Original version: Bjureberg et al., 2016; Chinese translation: Wang et al., 2021). The DERS-16 includes 16 items rated on a 5-point scale from 1 (almost never) to 5 (almost always). It evaluates five dimensions: lack of emotional clarity (Clarity; 2 items), difficulties engaging

in goal-directed behavior (Goal; 3 items), impulse control difficulties (Impulse; 3 items), limited access to effective emotion regulation strategies (Strategies; 5 items), and nonacceptance of emotional responses (Nonacceptance; 3 items). The DERS-16 has good reliability and validity. In the current sample, internal consistency of these subscales was acceptable: (1) Time 1: $\alpha s = 0.73–0.89$; (2) Time 2: $\alpha s = 0.74–0.88$.

Problematic TikTok Use

We adapted the Smartphone Addiction Scale–Short Version (SAS-SV; Original Version: Kwon et al., 2013; Chinese version: Chen et al., 2017) by replacing the word "smartphone" with "TikTok" to assess problematic TikTok use. The SAS-SV contains 10 items rated on a 6-point scale from 1 (strongly disagree) to 6 (strongly agree). The SAS-SV has good reliability and validity (Kwon et al., 2013). In the current sample, it showed good internal consistency: Time 1, $\alpha = 0.88$; Time 2, $\alpha = 0.89$.

Data Analysis

All analyses were conducted using R version 4.4.1 and RStudio 2024.04.2. Network analyses were performed using several R packages, including bootnet (Epskamp et al., 2018), glmnet (Friedman, et al., 2010), networktools (Jones, 2020), and qgraph (Epskamp et al., 2012).

Average scores on the DERS-16 subscales (i.e., Clarity; Goals; Impulse; Strategies; Nonacceptance) and item ratings on the adapted SAS-SV for PTU were used as node indicators (see Appendix 1 for node indicators). To reduce redundancy, conceptually overlapping items measuring PTU were combined: items 1 and 2 of the adapted SAS-SV (reflecting task interference; $r = 0.78$ at both assessments) and items 4 and 5 (reflecting withdrawal-like symptoms; $r = 0.78–0.79$ across assessments) were merged (Arrivillaga et al., 2023). We collapsed these nodes given that their strong association with each other could potentially suppress their associations with other nodes in the network (Wysocki et al., 2022).

Two contemporaneous networks were estimated each at Time 1 and Time 2, using the graphical least absolute shrinkage and selection operator and the extended Bayesian information criterion (EBICglasso) to select the optimal regularization parameter (Epskamp et al., 2012, 2018). Core pathways connecting PTU and DER were identified by examining the bridge edges (Jia et al., 2024). To identify the most influential nodes bridging PTU and DER, we calculated the two-step bridge expected influence (bridge-EI; Jones, 2020; Jones et al., 2021). Bridge EI1 reflects the sum of all direct associations between a node and nodes outside its community, while bridge EI2 accounts for the indirect effects a node exerts on other communities through intermediary nodes. Nodes with higher bridge EI1 and EI2 values have a stronger association with nodes in the other construct.

For the CLPN, we applied least absolute shrinkage and selection operator (LASSO) regularization with tenfold cross-validation (Wysocki et al., 2022; Zainal & Newman, 2023). We focused on bridge edges and their directional flow to explore pathways linking PTU and DER over time (Jia et al., 2024). Additionally, we calculated bridge EI1 and EI2, with EI in directed networks based solely on outgoing edges from each node (i.e., out-IE). Nodes with higher bridge EI1 and EI2 values have a stronger influence on nodes in the other construct.

Accuracy and stability of these networks were evaluated using nonparametric bootstrapping to estimate edge confidence intervals (CIs). Narrower CIs indicated greater accuracy of edge weights. Correlation stability (CS) coefficients for edge and bridge-EI were determined using case-drop bootstraps. CS coefficients above 0.25, 0.50, and 0.70 indicated acceptable, good, and excellent stability, respectively (Epskamp et al., 2018).

Results

Contemporaneous Network

Figure 1 presents the contemporaneous networks at Time 1 and Time 2. Stability and accuracy for these contemporaneous networks were supported. The CIs for edge weights in Time 1 and Time 2 networks are presented in Appendix 2. The case-dropping analysis (Appendix 3) demonstrated excellent stability for edge and bridge-EI at both time points, with CS coefficients of 0.75 for each metric.

As indicated in Appendix 4, the strongest bridge edge linking PTU and DER involves task interference due to TikTok use (i.e., missing planned work and difficulty concentrating on tasks due to TikTok use) and difficulties in goal-directed behavior at both time points. In terms of bridge-EI (Fig. 2), task interference due to TikTok use emerged as the node with the highest bridge-EI across both time points (T1: bridge EI1 = 0.41, bridge EI2 = 0.76; T2: bridge EI1 = 0.48, bridge EI2 = 0.81).

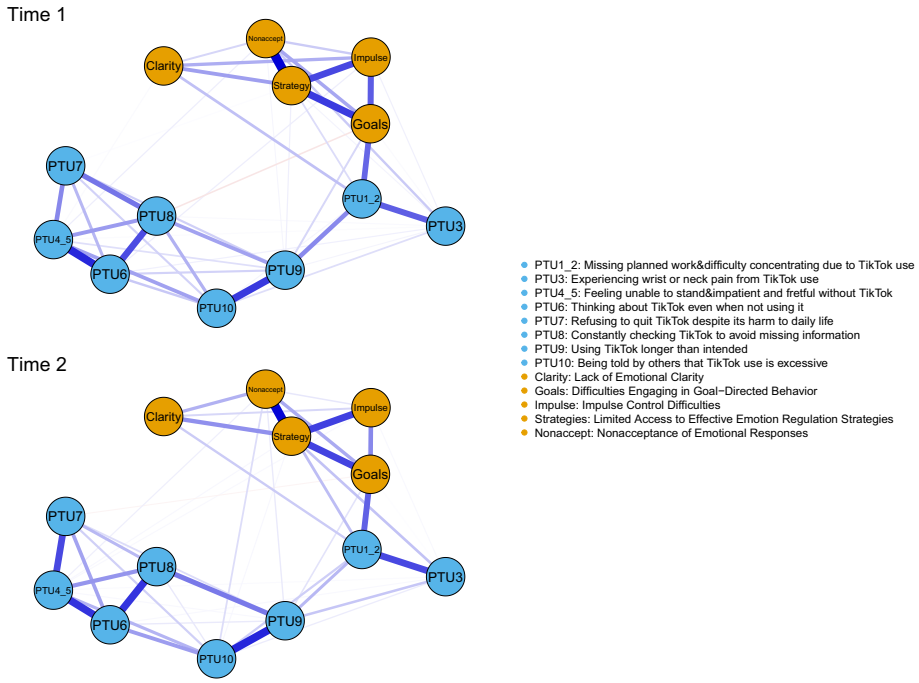

Fig. 1 Contemporaneous networks

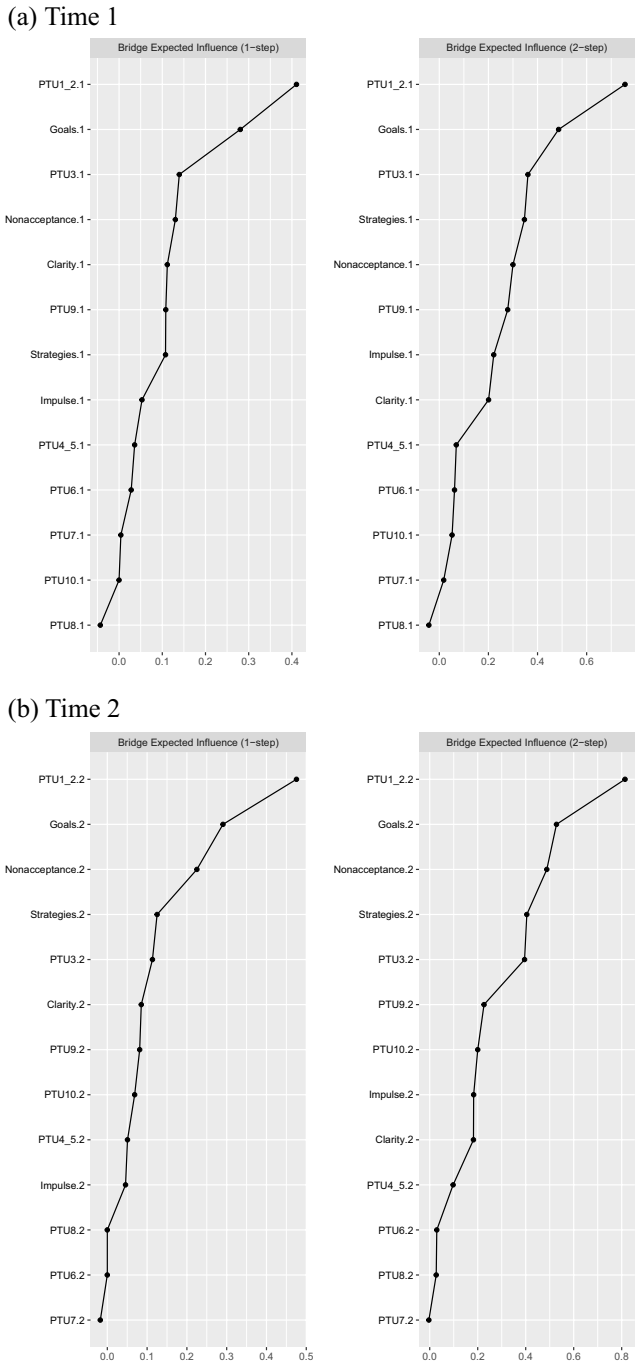

Fig. 2 Two-step bridge expected influence estimates at Time 1 and Time 2. *Note*: For the meaning of node names, see Appendix 1

Cross-lagged Panel Network

Figure 3 presents the CLPN. Stability and accuracy for the cross-lagged network were supported. CIs for edge weights in the CLPN are presented in Appendix 5. Case-dropping analyses (Appendix 6) showed stability was acceptable for bridge-EI (CS coefficient = 0.28) and edge (CS coefficient = 0.44).

As shown in Appendix 7, the strongest bridge edges connecting DER and PTU are from DER to PTU rather than the reverse direction: Impulse control difficulties → experiencing wrist or neck pain from TikTok use (value = 0.18), and difficulties with goal-directed behavior → using TikTok longer than intended (value = 0.17). Interestingly, a reciprocal relationship was identified between difficulties in goal-directed behavior and task interference due to TikTok use: task interference due to TikTok use → Goals (value = 0.13), and Goals → task interference due to TikTok use (value = 0.11). Counterintuitively, inverse associations were found between lack of emotional clarity and some PTU symptoms, which should be interpreted with caution.

Figure 4 illustrates the bridge out-EI1 and out-EI2. Difficulties with impulse control (out-EI1 = 0.43, out-EI2 = 0.77), goal-directed behavior (out-EI1 = 0.41, out-EI2 = 0.73), and task interference due to TikTok use (out-EI1 = 0.38, out-EI2 = 0.73) exhibited the highest bridge out-EI1/out-EI2 in the cross-lagged network.

Discussion

The current study employed a network analysis approach to gain a more nuanced understanding of how DER and PTU are associated over time. The findings address the key research questions by revealing the directionality of their relations, as well as identifying bridge nodes and edges connecting DER and PTU.

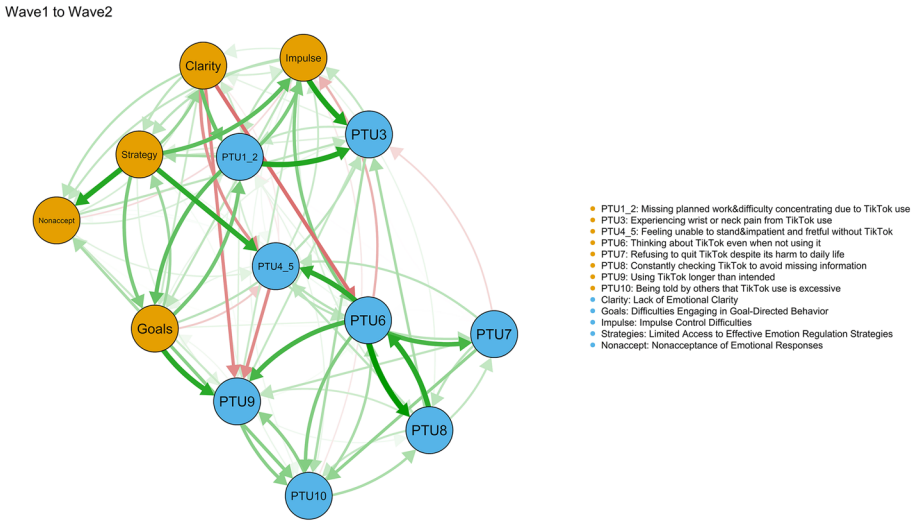

Fig. 3 Cross-lagged panel network. *Note*: Autoregressive edges were excluded from the figures. Green lines indicate positive associations, while red lines indicate negative associations. Line thickness and boldness signify strength of associations

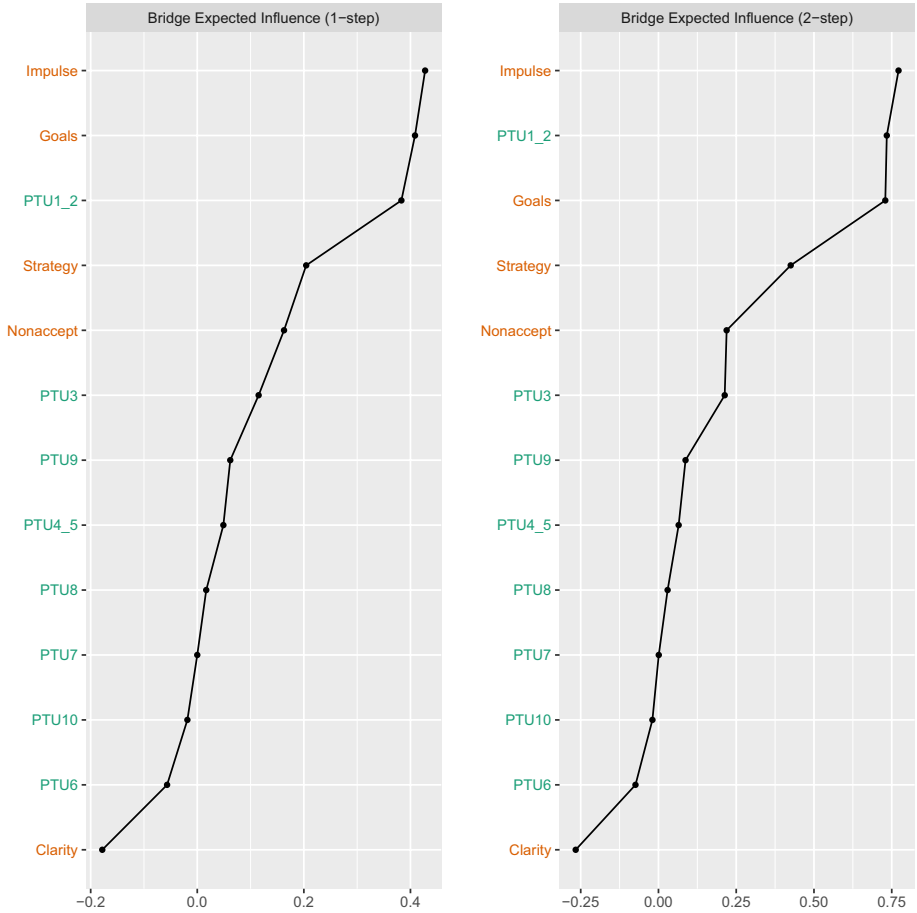

Fig. 4 Bridge out-expected influence derived from the cross-lagged model. *Note*: For the meaning of node names, see Appendix 1

Hypothesis Validation

Most of the strongest associations in the cross-lagged network flowed from DER to PTU, suggesting that nodes in the DER network may play a more dominant role in activating nodes in the PTU network. This pattern of results aligns with CIUT, which posits that DER may motivate individuals to use the Internet as a coping for emotional distress, thereby contributing to development of PIU (Kardefelt-Winther, 2014). Additionally, these findings are consistent with the I-PACE model (Brand et al., 2016, 2019), which conceptualizes DER as an affective and cognitive component contributing to PIU, although our study examines only a subset of the model's broader framework.

Examining the bridge edges linking DER and PTU, the two strongest associations were from impulse control difficulties to experiencing wrist or neck pain due to TikTok use, and from difficulties in goal-directed behavior to using TikTok longer than intended. Individuals with impulse control difficulties often indulge their impulses and seek immediate gratification to alleviate distress (Tice et al., 2001), while TikTok

readily provides immediate rewards. This may lead them to use TikTok as a means of emotion regulation, even to the point of physical discomfort. On the other hand, difficulty focusing on goals is associated with procrastination (Gustavson et al., 2015). In the context of TikTok use, these individuals may turn to the platform to procrastinate and escape from daily responsibilities when distressed. As it helps to distract from the pressure to pursue goals, the use of TikTok is reinforced, leading to prolonged use.

For the key bridge nodes, difficulties with impulse control and goal-directed behavior from the DER network had the highest bridge out-EI, indicating their strong influence on PTU within the cross-lagged network. These results align with previous research on DER dimensions in relation to PIU (Arrivillaga et al., 2023; Horwood & Anglim, 2021; Wartberg et al., 2021), which identified difficulties in goal-directed behavior and impulse control as key contributors to problematic use after controlling for the other DER dimensions.

Notably, missing planned work and difficulty concentrating on tasks due to TikTok use also showed a high bridge out-EI in the cross-lagged network, suggesting this node could exert effect on DER. Consistently, reciprocal relationships were found between task interference due to TikTok use and difficulties in goal-directed behavior. It is possible that when individuals struggle to focus on goals, they may turn to TikTok to procrastinate (Gustavson et al., 2015; Krause & Freund, 2014). However, TikTok use may exacerbate the issue by consuming time and hindering task completion, making it even harder to refocus on goals. Thus, both difficulty with goal-directed behavior and the use of TikTok at the expense of task interference are intensified. Nevertheless, it is important to acknowledge that some overlap exists between PTU items assessing task interference due to TikTok and DERS-16 items regarding difficulties in goal-directed behavior. Although designed to measure distinct constructs (i.e., daily-life disturbances due to TikTok use versus goal-directed behavior during emotional distress), these items both reflect impaired goal management, which may partly contribute to the observed cross-lagged relationships. To address this, future research could refine the measurement of task interference due to TikTok use by incorporating items minimizing overlap with difficulties in goal-directed behavior.

Meanwhile, the inverse association between lack of emotional clarity and PTU symptoms requires careful consideration, as this effect emerged after adjusting for effects of other DER facets and PTU symptoms at time 1 (see Appendix 8 for the bivariate correlations between all DER facets and PTU symptoms). Emotional clarity is often considered an initial step in effective emotion regulation, allowing individuals to accurately identify and interpret their emotional states. This understanding facilitates subsequent regulatory processes such as emotion acceptance, selecting appropriate coping strategies, impulse control, and maintaining goal-directed behavior (Eckland & Berenbaum, 2021; Vine & Aldao, 2014; Vine & Marroquín, 2018). Consistent with this framework, Vine and Aldao (2014) demonstrated that lack of clarity contributes to downstream difficulties in these processes, which subsequently predict various forms of psychopathology. For example, they found that lack of clarity increased impulse control difficulties, which in turn predicted problematic alcohol use. In the context of the present study, considering the role of emotional clarity in facilitating these regulatory processes, it is likely that lack of clarity primarily influences PTU symptoms indirectly through its effects on other DER facets. Consequently, after accounting for shared variance with these other facets, the residual variance of lack of clarity may have appeared to exert an inverse effect on PTU symptoms.

Contributions

This study makes several key contributions to the existing literature. First, we adopted a cross-lagged network analysis to test temporal precedence and bidirectional influences between DER facets and PTU symptoms over time. Our findings revealed that influences between DER facets and PTU symptoms predominantly flow from DER to PTU, while also showing paths from PTU to DER. Notably, difficulties in impulse control and goal-directed behavior were identified as central facets in associations between DER and PTU symptoms. These results address important gaps in the literature regarding the directionality of these constructs and offer a more precise understanding of their dynamic relationship. More broadly, given that TikTok use represents a specific form of Internet use, the current results also enhance our understanding of the bidirectional relationship between DER and PIU.

Second, the current findings have important clinical implications. A key clinical takeaway from our study is that focusing on specific facets of DER, particularly difficulties in impulse control and goal-directed behavior, may be an effective strategy for identifying individuals at risk for PTU. Our network analysis revealed that these two facets serve as prominent bridge nodes linking DER and PTU, with the strongest relationships emerging from these difficulties to specific PTU symptoms. This suggests that screening protocols for PTU may benefit from incorporating assessments of these DER facets.

In terms of treatment, our findings suggest that enhancing impulse control and promoting goal-directed behavior during distress could be promising targets for interventions aimed at reducing PTU. Although prior studies have explored emotion regulation interventions for PIU, this area remains relatively underexplored (Chrétien et al., 2025). Emotion-focused group therapy, aimed at reducing negative emotions and enhancing positive ones, has shown promise in alleviating PIU symptoms (Amini et al., 2020). Similarly, dialectical behavior therapy (DBT) skills training, including modules on mindfulness, distress tolerance, and emotion regulation, has yielded positive outcomes when adapted for PIU (El-Ashry et al., 2023; Pluhar et al., 2020). CBT-based group interventions that include emotion regulation skills have also been effective in reducing PIU severity (Lindenberg et al., 2022; Szász-Janocha et al., 2021). While these interventions suggest the potential of improving emotion regulation as a treatment approach for PIU, they do not specify which components of emotion regulation have the most substantial impact. Our study points to impulse control and goal-directed behavior as key intervention targets, given their central bridging role in the DER-PTU network.

It is important to note, however, that our study was not conducted in a treatment-seeking sample, so these clinical implications are preliminary. Future research is needed to determine whether interventions specifically targeting impulse control and goal-directed behavior during distress can effectively reduce PTU severity. Furthermore, optimal strategies for enhancing these facets remain unclear. It is possible that targeting other DER facets may indirectly strengthen impulse control and goal-directed behavior, resulting in broader treatment benefits. Further research should explore these possibilities and refine intervention strategies accordingly.

Limitations

The current study has limitations. First, the current study could only reflect short-term relations, instead of long-term influences, between DER and PTU. While our findings

highlight dynamic associations between these variables, future research with longer follow-up intervals and multiple time points could provide a deeper understanding of the stability and fluctuation of these relationships over time.

Second, we used the DERS-16, which does not have the dimension of lack of emotional awareness (Awareness) from the original DERS. However, prior studies have indicated that Awareness generally has weak correlations with other DER dimensions and is not strongly associated with criteria variables theoretically linked to DER; consequently, some researchers recommend excluding this dimension (Benfer et al., 2019; Hallion et al., 2018; Osborne et al., 2017). Future research could still benefit from incorporating the full DERS. Including the Awareness dimension may provide a more comprehensive understanding of DER's relationship with PTU.

Third, our sample was drawn from China and primarily consisted of individuals with relatively high educational attainment, recruited through an online platform. This limits generalizability of findings to other cultural and demographic contexts. Cultural factors may shape emotion regulation strategies and Internet use behaviors. For instance, individuals from collectivistic cultures are more likely to engage in emotion suppression than those from individualistic cultures (Huwaë & Schaafsma, 2018). Moreover, PIU severity in China has increased between 2003 and 2020, possibly due to rapid technological advancements and Internet availability outpacing the development of guidelines for healthy use (Han et al., 2024). Cross-cultural research has also highlighted both universal and culture-specific risk factors for PIU, with non-European countries exhibiting higher PIU severity (Lopez-Fernandez et al., 2023). Consequently, interpreting our results requires consideration of Chinese cultural norms and internet use behaviors.

Additionally, the relatively high educational attainment of our sample may have influenced the observed patterns of PTU. Prior research suggests that individuals with higher educational attainment are more likely to use the internet for studying and working, while differences in educational attainment have a lesser impact on internet use for social and entertainment purposes (Yu & Liu, 2022). This demographic skew may have had some influence on the observed DER-PTU relationships.

Future research could address these limitations by recruiting more diverse samples across cultural and educational backgrounds. Cross-cultural studies can help identify both shared and unique factors that shape the DER-PTU relationship, offering a more nuanced understanding of these associations. Additionally, meta-analyses synthesizing findings across different countries and demographic contexts could clarify the moderating effects of these factors, further enriching our knowledge of the complex interplay between DER, PTU, and contextual influences.

Conclusions

The current study offers a refined understanding of how DER dimensions and PTU symptoms relate over time. Our findings suggest that DER and PTU are dynamically linked, with bridge nodes such as difficulties with goal-directed behavior and impulse control in DER and missing planned work and difficulty concentration on tasks due to TikTok use in PTU, acting as key drivers of this relationship. Although stronger influences flowed from DER to PTU, a reciprocal relation was also observed between task interference due to TikTok use and Goals. Preventive strategies could focus on these bridge edges and nodes to mitigate problematic use effectively.

Appendix 1

Node names

- PTU1_2: Missing planned work and difficulty concentrating on tasks due to TikTok use
- PTU3: Experiencing wrist or neck pain from TikTok use
- PTU4_5: Feeling unable to stand and impatient and fretful without TikTok
- PTU6: Thinking about TikTok even when not using it
- PTU7: Refusing to quit TikTok despite its harm to daily life
- PTU8: Constantly checking TikTok to avoid missing information
- PTU9: Using TikTok longer than intended
- PTU10: Being told by others that TikTok use is excessive
- Clarity: Lack of Emotional Clarity
  - I have difficulty making sense out of my feelings
  - I am confused about how I feel
- Goals: Difficulties Engaging in Goal-Directed Behavior
  - When I am upset, I have difficulty getting work done
  - When I am upset, I have difficulty focusing on other things
  - When I am upset, I have difficulty thinking about anything else
- Impulse: Impulse Control Difficulties
  - When I am upset, I become out of control
  - When I am upset, I feel out of control
  - When I am upset, I have difficulty controlling my behaviors
- Strategies: Limited Access to Effective Emotion Regulation Strategies
  - When I am upset, I believe that I will remain that way for a long time
  - When I am upset, I believe that I'll end up feeling very depressed
  - When I am upset, I believe that there is nothing I can do to make myself feel better
  - When I am upset, I start to feel very bad about myself
  - When I am upset, my emotions feel overwhelming
- Nonacceptance: Nonacceptance of Emotional Responses.
  - When I am upset, I feel ashamed with myself for feeling that way
  - When I am upset, I feel like I am weak
  - When I am upset, I become irritated with myself for feeling that way

Note. PTU1_2 and PTU4_5 represent the collapsing of items 1 and 2 and items 4 and 5, respectively (see the "Data Analysis" section). The average scores of the emotion dysregulation facets were used as node indicators.

Appendix 2

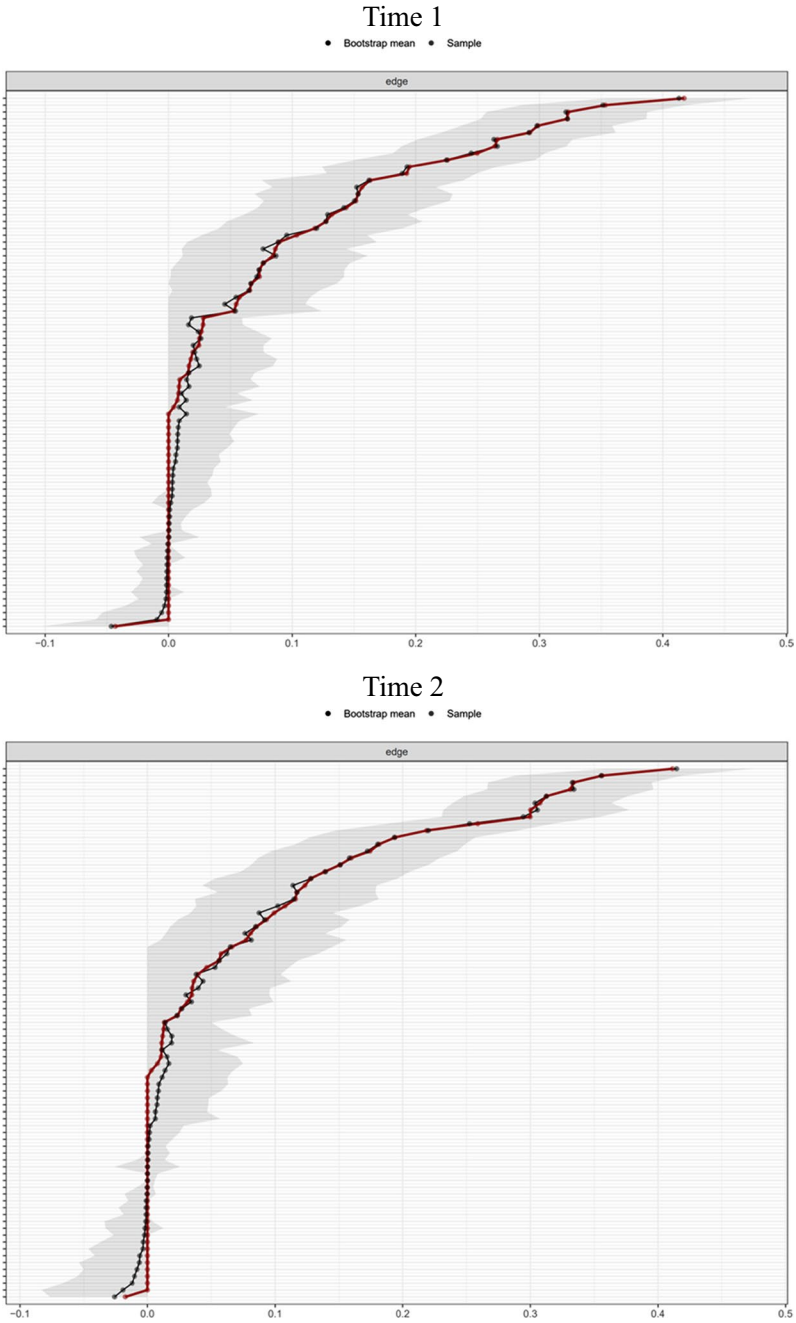

Fig. 5 Accuracy of edge estimates in the contemporaneous networks at Time 1 and Time 2. *Note*: The red point represents the sample value for each edge, while the black point indicates the bootstrap mean. The gray line depicts the bootstrapped confidence intervals

Appendix 3

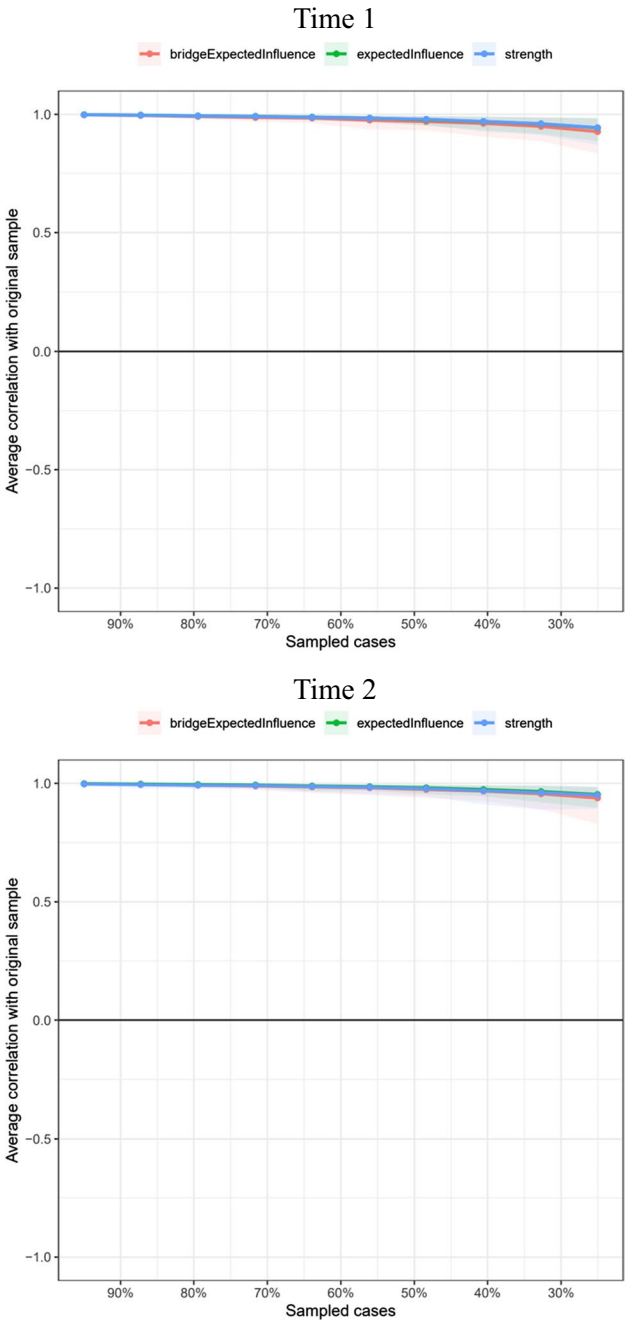

Fig. 6 Stability of centrality indices in the contemporaneous networks at Time 1 and Time 2. Note. The *x*-axis represents the percentage of cases removed from the original sample, and the *y*-axis shows the correlation between the remaining sample and the original sample. The lines indicate the mean values, while the shaded areas represent the 95% confidence interval

Appendix 4

Table 1 Edge weights in the contemporaneous networks

Time 1 Undirected edge	Weight	Time 2 Undirected edge	Weight
Strategies.1—Nonacceptance.1	0.417	Strategies.2—Nonacceptance.2	0.411
PTU4_5.1—PTU6.1	0.353	PTU9.2—PTU10.2	0.356
PTU9.1—PTU10.1	0.323	PTU4_5.2—PTU6.2	0.334
Goals.1—Strategies.1	0.323	PTU6.2—PTU8.2	0.331
Impulse.1—Strategies.1	0.299	Impulse.2—Strategies.2	0.313
PTU6.1—PTU8.1	0.292	Goals.2—Strategies.2	0.307
Goals.1—Impulse.1	0.266	PTU4_5.2—PTU7.2	0.300
PTU1_2.1—PTU3.1	0.264	PTU1_2.2—PTU3.2	0.300
PTU1_2.1—Goals.1	0.250	PTU1_2.2—Goals.2	0.259
PTU7.1—PTU8.1	0.226	PTU8.2—PTU9.2	0.219
PTU4_5.1—PTU7.1	0.195	Goals.2—Impulse.2	0.194
PTU1_2.1—PTU9.1	0.193	Clarity.2—Strategies.2	0.181
PTU4_5.1—PTU8.1	0.163	PTU4_5.2—PTU8.2	0.174
Clarity.1—Strategies.1	0.156	PTU6.2—PTU10.2	0.160
PTU4_5.1—PTU10.1	0.153	PTU6.2—PTU7.2	0.151
PTU8.1—PTU9.1	0.151	Goals.2—Nonacceptance.2	0.139
Goals.1—Nonacceptance.1	0.143	Clarity.2—Nonacceptance.2	0.128
Clarity.1—Impulse.1	0.132	PTU4_5.2—PTU10.2	0.123
PTU8.1—PTU10.1	0.127	PTU1_2.2—PTU9.2	0.117
PTU6.1—PTU7.1	0.118	PTU7.2—PTU8.2	0.116
PTU1_2.1—Clarity.1	0.104	PTU1_2.2—Nonacceptance.2	0.108

Only edges with a weight greater than 0.10 are presented. For the meaning of the node names, see Appendix 1.

Appendix 5

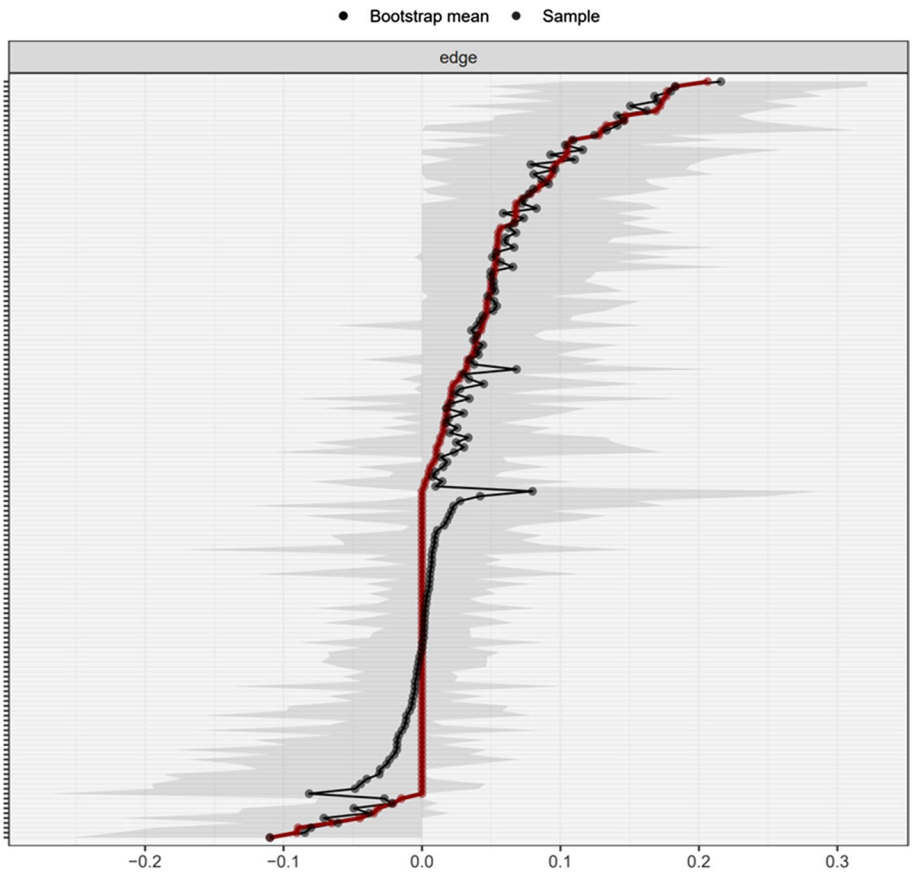

Fig. 7 Accuracy of the edge estimates for the cross-lagged panel network. *Note*: The red point on each line indicates the sample value of the edge, while the gray point shows the bootstrap mean. The gray lines represent the bootstrapped confidence intervals for each edge

Appendix 6

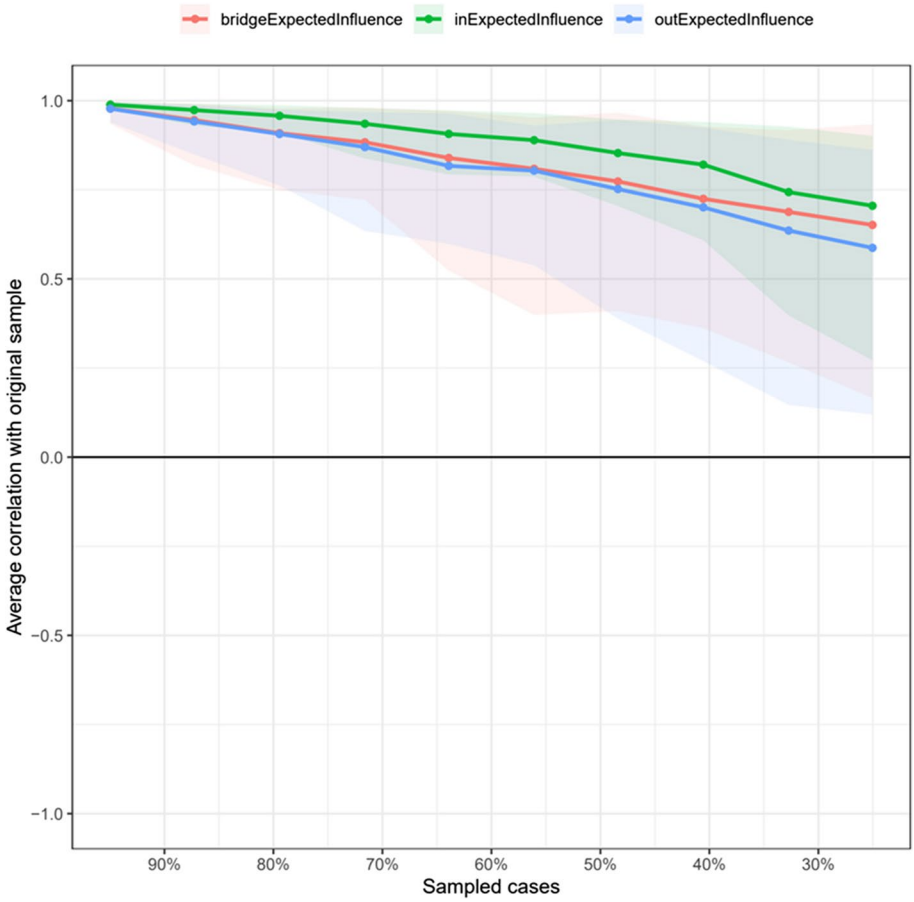

Fig. 8 Stability of centrality indices in the cross-lagged panel network. *Note*: The *x*-axis shows the percentage of cases removed from the original sample, while the *y*-axis reflects the correlation between the remaining sample and the original sample. The lines represent the mean values, and the shaded areas depict the 95% confidence interval

Appendix 7

Table 2 Non-zero bridge edges in the temporal network

nodeOut	nodeIn	value	DomainOut	DomainIn
Impulse	PTU3	0.183	DER	PTU
Goals	PTU9	0.172	DER	PTU
Strategy	PTU4_5	0.17	DER	PTU
PTU1_2	Goals	0.128	PTU	DER
Clarity	PTU1_2	0.12	DER	PTU
Goals	PTU1_2	0.111	DER	PTU
PTU1_2	Impulse	0.104	PTU	DER
Impulse	PTU6	0.098	DER	PTU
Goals	PTU10	0.095	DER	PTU
PTU1_2	Strategy	0.072	PTU	DER
Nonaccept	PTU10	0.054	DER	PTU
Goals	PTU3	0.054	DER	PTU
Nonaccept	PTU3	0.053	DER	PTU
Impulse	PTU1_2	0.051	DER	PTU
PTU3	Impulse	0.048	PTU	DER
Impulse	PTU9	0.044	DER	PTU
PTU9	Nonaccept	0.043	PTU	DER
PTU4_5	Impulse	0.042	PTU	DER
PTU1_2	Clarity	0.04	PTU	DER
PTU1_2	Nonaccept	0.039	PTU	DER
Nonaccept	PTU1_2	0.029	DER	PTU
Impulse	PTU8	0.029	DER	PTU
PTU3	Strategy	0.028	PTU	DER
Nonaccept	PTU4_5	0.024	DER	PTU
Strategy	PTU9	0.023	DER	PTU
PTU3	Clarity	0.021	PTU	DER
Goals	PTU6	0.018	DER	PTU
PTU3	Goals	0.018	PTU	DER
PTU4_5	Goals	0.018	PTU	DER
PTU8	Impulse	0.017	PTU	DER
Impulse	PTU10	0.013	DER	PTU
Strategy	PTU6	0.012	DER	PTU
PTU9	Goals	0.011	PTU	DER
Impulse	PTU4_5	0.009	DER	PTU
PTU9	Strategy	0.008	PTU	DER
PTU6	Nonaccept	0.005	PTU	DER
PTU4_5	Nonaccept	0.004	PTU	DER
Nonaccept	PTU6	0.003	DER	PTU
PTU4_5	Clarity	– 0.015	PTU	DER
PTU10	Impulse	– 0.019	PTU	DER
Goals	PTU4_5	– 0.042	DER	PTU
PTU6	Impulse	– 0.062	PTU	DER
Clarity	PTU4_5	– 0.089	DER	PTU
Clarity	PTU9	– 0.095	DER	PTU
Clarity	PTU6	– 0.115	DER	PTU

Only nonzero edges are presented. For the meaning of the node names, see Appendix 1. Abbreviations: *DER* difficulties in emotion regulation, *PTU* problematic TikTok use

Appendix 8

Table 3 Bivariate correlation between emotion dysregulation and problematic TikTok use

	clarity.1	goals.1	impulse.1	strategies.1	nonaccept.1	clarity.2	goals.2	impulse.2	strategies.2	nonaccept.2	PTU.1
clarity.1											
goals.1	0.47**										
impulse.1	0.53**	0.71**									
strategies.1	0.57**	0.79**	0.77**								
nonaccept.1	0.47**	0.67**	0.68**	0.78**							
clarity.2	0.57**	0.40**	0.43**	0.47**	0.41**						
goals.2	0.44**	0.76**	0.59**	0.66**	0.56**	0.45**					
impulse.2	0.42**	0.54**	0.65**	0.59**	0.48**	0.46**	0.65**				
strategies.2	0.48**	0.66**	0.62**	0.74**	0.60**	0.56**	0.75**	0.71**			
nonaccept.2	0.41**	0.56**	0.53**	0.61**	0.60**	0.51**	0.65**	0.60**	0.77**		
PTU.1	0.32**	0.45**	0.40**	0.45**	0.42**	0.26**	0.45**	0.36**	0.43**	0.40**	
PTU.2	0.27**	0.42**	0.40**	0.43**	0.39**	0.31**	0.48**	0.39**	0.49**	0.49**	0.75**

0.1 = Time 1; 0.2 = Time 2. CLARITY = lack of emotional clarity; GOALS = difficulties engaging in goal-directed behavior; IMPULSE = impulse control difficulties; STRATEGIES = limited access to effective emotion regulation strategies; NONACCEPT = nonacceptance of emotional responses; PTU = problematic TikTok use

Author Contributions N. Yao: conceptualization, formal analysis, writing—original draft, writing—review and editing; J. Chen: conceptualization, validation, writing—original draft, writing—review and editing; J. D. Elhai: conceptualization, writing—review and editing.

Data Availability Data will be made available on request.

Declarations

Ethical Approval and Consent The study procedures were conducted in accordance with the Declaration of Helsinki and approved by the ethics committee of the first author's university (ID: 2022A21). All participants were informed about the study and provided their consent to participate.

Competing interests N. Yao, J. Chen, and J. D. Elhai declare that they have no conflict of interest. For reasons of transparency, Jon Elhai notes that he receives royalties for several books published on posttraumatic stress disorder (PTSD); is a paid, full-time faculty member at University of Toledo; occasionally serves as a paid, expert witness on PTSD legal cases; and recently received grant research funding from the US National Institutes of Health.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process The authors used generative AI (ChatGPT-4, OpenAI Inc., https://openai.com/) to enhance the readability and language of this article. Following the use of the tool, the authors carefully reviewed and edited the content to ensure its accuracy and quality, and they take full responsibility for the final published material.

References

Akbari, M. (2017). Metacognitions or distress intolerance: The mediating role in the relationship between emotional dysregulation and problematic internet use. *Addictive Behaviors Reports, 6*, 128–133. https://doi.org/10.1016/j.abrep.2017.10.004

Amendola, S., Spensieri, V., Guidetti, V., & Cerutti, R. (2019). The relationship between difficulties in emotion regulation and dysfunctional technology use among adolescents. *Journal of Psychopathology*, *25*(1), 10–17. https://hdl.handle.net/11573/1231956

Amini, M., Lotfi, M., Fatemitabar, R., & Bahrampoori, L. (2020). The effectiveness of emotion-focused group therapy on the reduction of negative emotions and internet addiction symptoms. *Practice in Clinical Psychology*, *8*(1), 1–8. https://doi.org/10.32598/jpcp.8.1.1

Arrivillaga, C., Hallauer, C. J., Montag, C., & Elhai, J. D. (2023). Emotion dysregulation factors associated with problematic smartphone use severity: The mediating role of fear of missing out. *Addictive Behaviors, 143*, Article 107708. https://doi.org/10.1016/j.addbeh.2023.107708

Aslam, S. (2024). TikTok by the numbers: Stats, demographics & fun facts. Omnicore. https://www.omnicoreagency.com/tiktok-statistics/. Accessed 5 Oct 2024

Benfer, N., Bardeen, J. R., Fergus, T. A., & Rogers, T. A. (2019). Factor structure and incremental validity of the original and modified versions of the difficulties in emotion regulation scale. *Journal of Personality Assessment, 101*(6), 598–608. https://doi.org/10.1080/00223891.2018.1492927

Bjureberg, J., Ljótsson, B., Tull, M. T., Hedman, E., Sahlin, H., Lundh, L. G., ... & Gratz, K. L. (2016). Development and validation of a brief version of the difficulties in emotion regulation scale: The DERS-16. *Journal of Psychopathology and Behavioral Assessment*, *38*, 284–296. https://doi.org/10.1007/s10862-015-9514-x

Borsboom, D. (2017). A network theory of mental disorders. *World Psychiatry, 16*(1), 5–13. https://doi.org/10.1002/wps.20375

Brand, M., Young, K. S., Laier, C., Wölfling, K., & Potenza, M. N. (2016). Integrating psychological and neurobiological considerations regarding the development and maintenance of specific Internet-use disorders: An Interaction of Person-Affect-Cognition-Execution (I-PACE) model. *Neuroscience and Biobehavioral Reviews, 71*, 252–266. https://doi.org/10.1016/j.neubiorev.2016.08.033

Brand, M., Wegmann, E., Stark, R., Müller, A., Wölfling, K., Robbins, T. W., & Potenza, M. N. (2019). The Interaction of Person-Affect-Cognition-Execution (I-PACE) model for addictive behaviors: Update, generalization to addictive behaviors beyond internet-use disorders, and specification of the process

character of addictive behaviors. *Neuroscience and Biobehavioral Reviews, 104*, 1–10. https://doi.org/10.1016/j.neubiorev.2019.06.032

Casale, S., Caplan, S. E., & Fioravanti, G. (2016). Positive metacognitions about Internet use: The mediating role in the relationship between emotional dysregulation and problematic use. *Addictive Behaviors, 59*, 84–88. https://doi.org/10.1016/j.addbeh.2016.03.014

Chen, B., Liu, F., Ding, S., Ying, X., Wang, L., & Wen, Y. (2017). Gender differences in factors associated with smartphone addiction: A cross-sectional study among medical college students. *BMC Psychiatry, 17*(1), 341. https://doi.org/10.1186/s12888-017-1503-z

Chrétien, S., Giroux, I., Smith, I., Jacques, C., Ferland, F., Sévigny, S., & Bouchard, S. (2025). Emotional regulation in substance-related and addictive disorders treatment: A systematic review. *Journal of Gambling Studies*, 1–96. https://doi.org/10.1007/s10899-024-10366-8

Donald, J. N., Ciarrochi, J., & Sahdra, B. K. (2022). The consequences of compulsion: A 4-year longitudinal study of compulsive internet use and emotion regulation difficulties. *Emotion (Washington, D.C.)*, *22*(4), 678–689. https://doi.org/10.1037/emo0000769

Eckland, N. S., & Berenbaum, H. (2021). Emotional awareness in daily life: Exploring its potential role in repetitive thinking and healthy coping. *Behavior Therapy, 52*(2), 338–349. https://doi.org/10.1016/j.beth.2020.04.010

El-Ashry, A. M., Hussein Ramadan Atta, M., Alsenany, S. A., Farghaly Abdelaliem, S. M., & Abdelwahab Khedr, M. (2023). The effect of distress tolerance training on problematic internet use and psychological wellbeing among faculty nursing students: A randomized control trial. *Psychology Research and Behavior Management*, 4015–4032. https://doi.org/10.2147/prbm.s423194

Epskamp, S., Borsboom, D., & Fried, E. I. (2018). Estimating psychological networks and their accuracy: A tutorial paper. *Behavior Research Methods, 50*, 195–212. https://doi.org/10.3758/s13428-017-0862-1

Epskamp, S., Cramer, A. O., Waldorp, L. J., Schmittmann, V. D., & Borsboom, D. (2012). qgraph: Network visualizations of relationships in psychometric data. *Journal of Statistical Software*, *48*(4), 1–18. https://doi.org/10.18637/jss.v048.i04

Faghani, N., Akbari, M., Hasani, J., & Marino, C. (2020). An emotional and cognitive model of problematic Internet use among college students: The full mediating role of cognitive factors. *Addictive Behaviors, 105*, Article 106252. https://doi.org/10.1016/j.addbeh.2019.106252

Friedman, J. H., Hastie, T., & Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, *33*(1), 1–22. https://doi.org/10.18637/jss.v033.i01

Gioia, F., Rega, V., & Boursier, V. (2021). Problematic internet use and emotional dysregulation among young people: A literature review. *Clinical Neuropsychiatry: Journal of Treatment Evaluation*, *18*(1), 41–54. https://doi.org/10.36131/cnfioritieditore20210104

Gratz, K. L., & Roemer, L. (2004). Multidimensional assessment of emotion regulation and dysregulation: Development, factor structure, and initial validation of the difficulties in emotion regulation scale. *Journal of Psychopathology and Behavioral Assessment, 26*, 41–54. https://doi.org/10.1023/B:JOBA.0000007455.08539.94

Gustavson, D. E., Miyake, A., Hewitt, J. K., & Friedman, N. P. (2015). Understanding the cognitive and genetic underpinnings of procrastination: Evidence for shared genetic influences with goal management and executive function abilities. *Journal of Experimental Psychology: General, 144*(6), 1063–1079. https://doi.org/10.1037/xge0000110

Hallion, L. S., Steinman, S. A., Tolin, D. F., & Diefenbach, G. J. (2018). Psychometric properties of the Difficulties in Emotion Regulation Scale (DERS) and its short forms in adults with emotional disorders. *Frontiers in Psychology, 9*, 539. https://doi.org/10.3389/fpsyg.2018.00539

Han, X., Zhang, S., Miao, S., Wu, Z., Potenza, M. N., & Su, W. (2024). Will Internet availability fuel problematic Internet use? A cross-temporal meta-analysis of patterns in China from 2003 to 2020. *Current Addiction Reports, 11*(1), 113–124. https://doi.org/10.1007/s40429-023-00537-9

Hew, Y. C., & Chew, P. K. (2025). Distress tolerance as a risk factor for specific Internet-use disorders. *Journal of Technology in Behavioral Science*, 1–12. https://doi.org/10.1007/s41347-025-00489-z

Horwood, S., & Anglim, J. (2021). Emotion regulation difficulties, personality, and problematic smartphone use. *Cyberpsychology, Behavior, and Social Networking, 24*(4), 275–281. https://doi.org/10.1089/cyber.2020.0328

Huwaë, S., & Schaafsma, J. (2018). Cross-cultural differences in emotion suppression in everyday interactions. *International Journal of Psychology, 53*(3), 176–183. https://doi.org/10.1002/ijop.12283

Jia, J., Tong, W., Wang, X., & Fang, X. (2024). The comorbidity mechanism of problematic internet use and depression among Chinese college students: A cross-lagged panel network analysis. *Addictive Behaviors, 156*, Article 108057. https://doi.org/10.1016/j.addbeh.2024.108057

Jones, P. J., Ma, R., & McNally, R. J. (2021). Bridge centrality: A network approach to understanding comorbidity. *Multivariate Behavioral Research, 56*(2), 353–367. https://doi.org/10.1080/00273171.2019.1614898

Jones, P. (2020). networktools: Tools for identifying important nodes in networks. R package version 1.5.2. https://CRAN.R-project.org/package=networktools. Accessed 5 Oct 2024

Kardefelt-Winther, D. (2014). A conceptual and methodological critique of internet addiction research: Towards a model of compensatory internet use. *Computers in Human Behavior, 31*(1), 351–354. https://doi.org/10.1016/j.chb.2013.10.059

Krause, K., & Freund, A. M. (2014). How to beat procrastination: The role of goal focus. *European Psychologist, 19*, 132–144. https://doi.org/10.1027/1016-9040/a000153

Kwon, M., Kim, D., Cho, H., & Yang, S. (2013). The smartphone addiction scale: Development and validation of a short version for adolescents. *PLoS ONE, 8*(12), e83558–e83558. https://doi.org/10.1371/journal.pone.0083558

Lindenberg, K., Kindt, S., & Szász-Janocha, C. (2022). Effectiveness of cognitive behavioral therapy–based intervention in preventing gaming disorder and unspecified internet use disorder in adolescents: A cluster randomized clinical trial. *JAMA Network Open, 5*(2), e2148995–e2148995. https://doi.org/10.1001/jamanetworkopen.2021.48995

Lopez-Fernandez, O., Romo, L., Kern, L., Rousseau, A., Lelonek-Kuleta, B., Chwaszcz, J., Männikkö, N., Rumpf, H. J., Bischof, A., Király, O., Gässler, A. K., Graziani, P., Kääriäinen, M., Landrø, N. I., Zacarés, J. J., Chóliz, M., Dufour, M., Rochat, L., Zullino, D., Achab, S., … Kuss, D. J. (2023). Problematic Internet use among adults: A cross-cultural study in 15 countries. *Journal of Clinical Medicine*, *12*(3), 1027. https://doi.org/10.3390/jcm12031027

Montag, C., Yang, H., & Elhai, J. D. (2021). On the psychology of TikTok use: A first glimpse from empirical findings. *Frontiers in Public Health, 9*, Article 641673. https://doi.org/10.3389/fpubh.2021.641673

Nong, W., He, Z., Ye, J. H., Wu, Y. F., Wu, Y. T., Ye, J. N., & Sun, Y. (2023). The relationship between short video flow, addiction, serendipity, and achievement motivation among Chinese vocational school students: The post-epidemic era Context. *Healthcare (Basel, Switzerland), 11*(4), 462. https://doi.org/10.3390/healthcare11040462

Osborne, T. L., Michonski, J., Sayrs, J., Welch, S. S., & Anderson, L. K. (2017). Factor structure of the difficulties in emotion regulation scale (DERS) in adult outpatients receiving dialectical behavior therapy (DBT). *Journal of Psychopathology and Behavioral Assessment, 39*, 355–371. https://doi.org/10.1007/s10862-017-9586-x

Paulus, F. W., Hübler, K., Mink, F., & Möhler, E. (2021). Emotional dysregulation in preschool age predicts later media use and gaming disorder symptoms in childhood. *Frontiers in Psychiatry*, *12*, Article 626387. https://doi.org/10.3389/fpsyt.2021.626387

Pluhar, E., Jhe, G., Tsappis, M., Bickham, D., & Rich, M. (2020). Adapting dialectical behavior therapy for treating problematic interactive media use. *Journal of Psychiatric Practice, 26*(1), 63–70. https://doi.org/10.1097/PRA.0000000000000439

Schettler, L. M., Thomasius, R., & Paschke, K. (2024). Emotional dysregulation predicts problematic gaming in children and youths: A cross-sectional and longitudinal approach. *European Child & Adolescent Psychiatry, 33*(2), 605–616. https://doi.org/10.1007/s00787-023-02184-x

Smith, T., & Short, A. (2022). Needs affordance as a key factor in likelihood of problematic social media use: Validation, latent profile analysis and comparison of TikTok and Facebook problematic use measures. *Addictive Behaviors, 129*, Article 107259. https://doi.org/10.1016/j.addbeh.2022.107259

Szász-Janocha, C., Vonderlin, E., & Lindenberg, K. (2021). Treatment outcomes of a CBT-based group intervention for adolescents with Internet use disorders. *Journal of Behavioral Addictions, 9*(4), 978–989. https://doi.org/10.1556/2006.2020.00089

Tice, D. M., Bratslavsky, E., & Baumeister, R. F. (2001). Emotional distress regulation takes precedence over impulse control: If you feel bad, do it! *Journal of Personality and Social Psychology, 80*(1), 53–67.

Tsai, J. K., Lu, W. H., Hsiao, R. C., Hu, H. F., & Yen, C. F. (2020). Relationship between difficulty in emotion regulation and internet addiction in college students: A one-year prospective study. *International Journal of Environmental Research and Public Health, 17*(13), 4766. https://doi.org/10.3390/ijerph17134766

Vine, V., & Aldao, A. (2014). Impaired emotional clarity and psychopathology: A transdiagnostic deficit with symptom-specific pathways through emotion regulation. *Journal of Social and Clinical Psychology, 33*(4), 319–342. https://doi.org/10.1521/jscp.2014.33.4.319

Vine, V., & Marroquín, B. (2018). Affect intensity moderates the association of emotional clarity with emotion regulation and depressive symptoms in unselected and treatment-seeking samples. *Cognitive Therapy and Research, 42*(1), 1–15. https://doi.org/10.1007/s10608-017-9870-9

Wang, J., Wang, M., & Lei, L. (2023). Longitudinal links among paternal and maternal harsh parenting, adolescent emotional dysregulation and short-form video addiction. *Child Abuse & Neglect, 141*, Article 106236. https://doi.org/10.1016/j.chiabu.2023.106236

Wang, G.M., Guo, W., Shen, J.J. (2021). Validity and reliability of the brief version of the difficulties in emotion regulation scale in Chinese college students. *Chinese Journal of Clinical Psychology*, *29*, 956–961. https://link.cnki.net/doi/https://doi.org/10.16128/j.cnki.1005-3611.2021.05.013

Wartberg, L., & Lindenberg, K. (2020). Predictors of spontaneous remission of problematic internet use in adolescence: A one-year follow-up study. *International Journal of Environmental Research and Public Health, 17*(2), 448. https://doi.org/10.3390/ijerph17020448

Wartberg, L., Thomasius, R., & Paschke, K. (2021). The relevance of emotion regulation, procrastination, and perceived stress for problematic social media use in a representative sample of children and adolescents. *Computers in Human Behavior, 121*, Article 106788. https://doi.org/10.1016/j.chb.2021.106788

Wysocki, A., van Bork, R., Cramer, A. O. J., & Rhemtulla, M. (2022). Cross-lagged network models. PsyArXiv. https://doi.org/10.31234/osf.io/vjr8z

Yang, J., Ti, Y., & Ye, Y. (2022a). Offline and online social support and short-form video addiction among Chinese adolescents: The mediating role of emotion suppression and relatedness needs. *Cyberpsychology, Behavior, and Social Networking, 25*(5), 316–322. https://doi.org/10.1089/cyber.2021.0323

Yang, H., Wang, Z., Elhai, J. D., & Montag, C. (2022b). The relationship between adolescent emotion dysregulation and problematic technology use: Systematic review of the empirical literature. *Journal of Behavioral Addictions, 11*(2), 290–304. https://doi.org/10.1556/2006.2022.00038

Ye, J. H., He, Z., Yang, X., Lee, Y. S., Nong, W., Ye, J. N., & Wang, C. L. (2023a). Predicting the learning avoidance motivation, learning commitment, and silent classroom behavior of Chinese vocational college students caused by short video addiction. *Healthcare (Basel, Switzerland), 11*(7), 985. https://doi.org/10.3390/healthcare11070985

Ye, J. H., Wu, Y. F., Nong, W., Wu, Y. T., Ye, J. N., & Sun, Y. (2023b). The association of short-video problematic use, learning engagement, and perceived learning ineffectiveness among Chinese vocational students. *Healthcare (Basel, Switzerland), 11*(2), 161. https://doi.org/10.3390/healthcare11020161

Ye, J. H., Chen, M. Y., & Wu, Y. F. (2024a). The causes, counseling, and prevention strategies for maladaptive and deviant behaviors in schools. *Behavioral Sciences (Basel, Switzerland), 14*(2), 118. https://doi.org/10.3390/bs14020118

Ye, J. H., Cui, Y., Wang, L., & Ye, J. N. (2024). The relationships between the short video addiction, self-regulated learning, and learning well-being of Chinese undergraduate students. *International Journal of Mental Health Promotion*, *26*(10). https://doi.org/10.32604/ijmhp.2024.055814

Yu, X., & Liu, S. (2022). Disparities in online use behaviours and Chinese digital inclusion: A 10-year comparison. *International Journal of Environmental Research and Public Health, 19*(19), 11937. https://doi.org/10.3390/ijerph191911937

Zainal, N. H., & Newman, M. G. (2023). A cross-lagged prospective network analysis of depression and anxiety and cognitive functioning components in midlife community adult women. *Psychological Medicine, 53*, 4160–4171. https://doi.org/10.1017/S0033291722000848

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.